

KAFALACARBON: AN INSURTECH-BASED ACTUARIAL PRICING FRAMEWORK FOR PARAMETRIC TAKAFUL PROTECTING CARBON CREDIT SELLERS AND ENHANCING CARBON MARKET RESILIENCE IN INDONESIA

Hesti Indriani¹, Gading Surya Permana¹ and Ruhiyat^{1*}

1 Department of Actuarial Science, School of Data Science Mathematics and Informatics, IPB University, Bogor, West Java, 16680, Indonesia;

*Corresponding author: ruhiyat-mat@apps.ipb.ac.id

THEME: Circular Economy

ABSTRACT

Indonesia's voluntary carbon market remains severely underdeveloped, with trading volume on IDXCarbon representing only 0.00064% of total supply, primarily due to reversal risk income loss caused by wildfire-induced invalidation of carbon credits for which no insurance mechanism currently exists. This study introduces KafalaCarbon, a parametric takaful framework designed to protect carbon credit sellers in Indonesia against wildfire-induced revenue losses while enhancing overall carbon market resilience. Using 13 years of NASA VIIRS satellite fire data from East Kalimantan (2012–2024), fire events were reconstructed via DBSCAN spatial clustering and calibrated against Global Forest Watch verified forest loss data. Fire frequency was modeled using a Negative Binomial distribution, and loss severity using a Lognormal distribution, selected through Akaike Information Criterion comparison. Actuarial premiums were derived on a $0.1^\circ \times 0.1^\circ$ spatial grid under the Sharia-compliant Wakalah bil Ujrah and Tabarru' contract structure. Results indicate severe overdispersion in annual fire counts (dispersion $index = 414.6$), with frequency parameters of $\lambda = 1,177$ events/year under normal conditions and $\lambda = 3,110$ events/year during El Niño years. Gross premiums range from 0.54% to 1.20% of net carbon revenue across seven market segments, well below conventional insurance benchmarks, yielding a coverage ratio of $83 \times$ at the recommended 2,500 – *hectare* cooperative entry point. KafalaCarbon is demonstrated to be actuarially sound, Sharia-compliant under DSN-MUI Fatwa No. 52 and 53 (2006), and operationally viable as a mobile-first InsurTech platform, offering a replicable model for parametric Islamic insurance covering environmental asset classes.

KEYWORDS: Parametric takaful; carbon credit insurance; wildfire risk; actuarial pricing; Sharia compliance